



Python for Data Science - 2305CS303

Lab - 2

Roll No. : 122

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1. WAP to check whether the given number is Positive or Negative.

```
In [2]: a=int(input("enter a: "))
if a<0:
    print("number is negative")
else:
    print("number is positive")
```

number is positive

2. WAP to check whether the given number is Odd or Even.

```
In [3]: a=int(input("enter a: "))
if a%2==0:
    print("number is even")
else:
    print("number is odd")
```

number is even

3. WAP to find out Largest number from given two numbers using simple if and ternary operator.

```
In [10]: a=int(input("enter a: "))
b=int(input("enter b: "))
if a>b:
    print("A is greater")
else:
```

```
print("B is greater")
c = a if a > b else b
print(c)
```

B is greater
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4. WAP to find out Largest number from given three numbers.

```
In [14]: a=int(input("enter a: "))
b=int(input("enter b: "))
c=int(input("enter c: "))
if a>b and a>c:
    print("a is greater")
elif b>a and b>c:
    print("b is greater")
else:
    print("c is greater")
```

5. WAP to check whether the given year is Leap year or not.

[If a year can be divisible by 4 but not divisible by 100 then it is leap year but if it is divisible by 400 then it is leap year].

```
In [18]: y=int(input("enter year"))
if (y%4==0 and y%100!=0) or y%400==0:
    print("leap year")
else:
    print("not leap year")
```

6. WAP to display the name of the Day according to the number given by the user.

```
In [22]: day_number = int(input("Enter a day number (1-7): "))

days = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]

if 1 <= day_number <= 7:
    print("The day is:", days[day_number - 1])
else:
    print("Invalid input! Please enter a number between 1 and 7.")
```

The day is: Monday

7. WAP to implement simple Calculator which performs (add,sub,mul,div) of two numbers based on user input.

```
In [21]: a=int(input("enter a: "))
b=int(input("enter b: "))
c=int(input("enter 1 for + , 2 for -, 3 for /, 4 for *"))
if c==1:
```

```

    print(a+b)
elif c==2:
    print(a-b)
elif c==3:
    print(a/b)
elif c==4:
    print(a*b)
else:
    print("enter between 1 to 4")

```

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8. WAP to calculate electricity bill based on following criteria. Which takes the unit from the user.

- First 1 to 50 units – Rs. 2.60/unit
- Next 50 to 100 units – Rs. 3.25/unit
- Next 100 to 200 units – Rs. 5.26/unit
- 200 units – Rs. 8.45/unit

```

In [20]: a=int(input("enter bill: "))
d=0
if a<=50:
    print(a*2.60)
elif a>50 and a<=100:
    print((50*2.60)+((a-50)*3.25))
elif a>100 and a<=200:
    print((50*2.60)+(50*3.25)+((a-100)*5.26))
else:
    print((50*2.60)+(50*3.25)+(50*5.26)+((a-150)*8.45))

```

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